

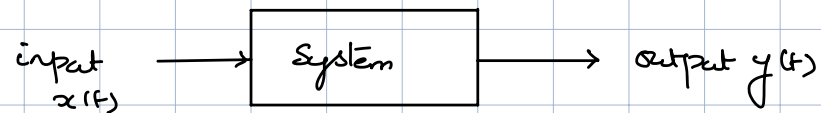
Signal:- A function that carries some information.

independent variable $\rightarrow t$

System:- Transforms a signal and generates another signal
↓ ↓
input signal $x(t)$ output signal $y(t)$

input $\rightarrow x(t)$

output $\rightarrow y(t)$



Mathematically, a system is defined by an input-output relation.

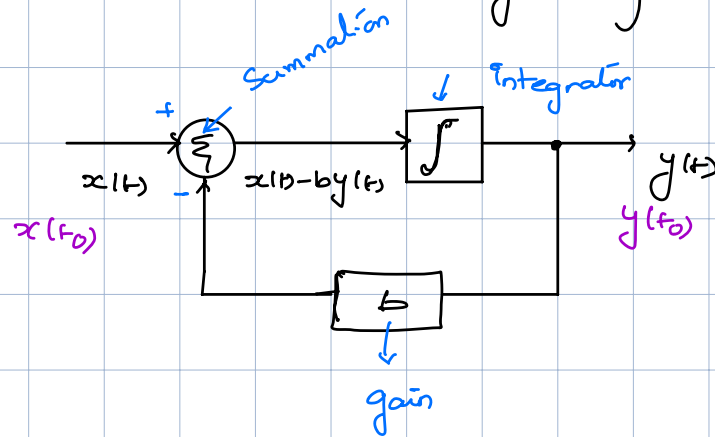
In this Course:- i/p - o/p relation is governed by a DE
(single ODE)
(system of DEs)

finding the response or o/p of a system $\Rightarrow$ Solving the DE

Example:- A system whose i/p - o/p relation $\frac{dy}{dt} + ay = x(t)$

$x \leftarrow$ input
 $y \leftarrow$ output

$$y(t) = \int x(t) - ay(t)$$

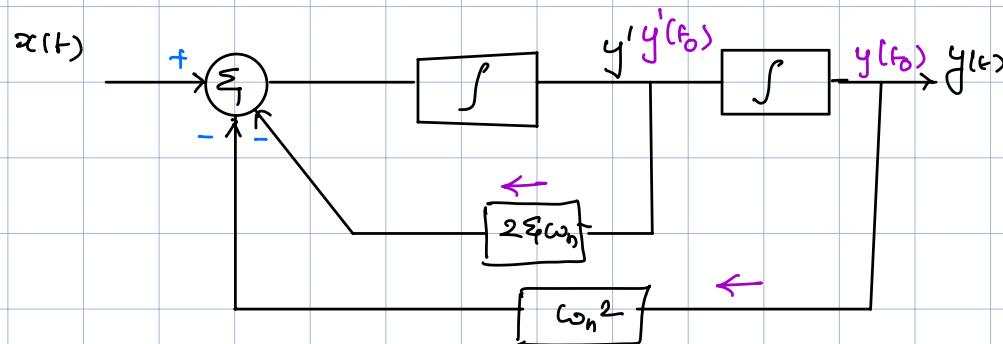


Systems whose i/p - o/p relation is governed by a DE } Basic building block is an integrator

Example:-

i/p - o/p Relation: $\frac{d^2y}{dt^2} + 2\xi\omega_n \frac{dy}{dt} + \omega_n^2 y = x(t)$

$$\hookrightarrow y'' = x(t) - 2\xi\omega_n y' - \omega_n^2 y$$



Linear System:-

$$\text{if } \begin{aligned} x_1(t) &\rightarrow y_1(t) \\ x_2(t) &\rightarrow y_2(t) \end{aligned}$$

then a system is linear if for the input $\alpha x_1(t) + \beta x_2(t)$
the o/p is $\alpha y_1(t) + \beta y_2(t)$. $\forall \alpha, \beta \in \mathbb{C}$ } Superposition

a) additivity:

$$\text{if } \begin{aligned} x_1(t) &\rightarrow y_1(t) \\ x_2(t) &\rightarrow y_2(t) \end{aligned}$$

Example:-

$$\sum_{i=1}^n a_i(t) y^{(i)}(t) + a_0(t) y(t) = x(t)$$

$$\Rightarrow L[y] = x(t)$$

$a_i(t) \rightarrow$ polynomial in t .

Linear -

Proof:-

$$y_1 = y: L[y] = x_1(t)$$

$$y_2 = y: L[y] = x_2(t)$$

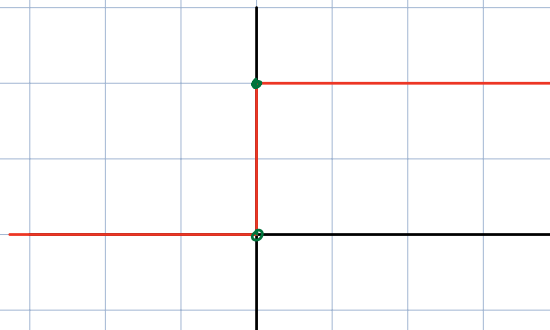
To show: if i/p is $\alpha x_1 + \beta x_2$, the o/p is

$$\alpha y_1 + \beta y_2$$

$$L[\alpha y_1 + \beta y_2] = \alpha x_1 + \beta x_2.$$

Step signal:- (Sorten)

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$



Step response of a system:-

↳

$$x(t) = u(t - \tau)$$

↓ System whose i/p-o/p relation is ODE

init cond: $t_0 < \tau$.

Solving $L[y] = u(t - \tau)$

$$y(t) = \begin{cases} y_1(t) & \forall t < \tau. \\ y_2(t) & \forall t \geq \tau. \end{cases}$$

↓

Continuum

↓

$$y_2(\tau) = y_1(\tau)$$

where $y_1(t) \rightarrow$ Sol of $L[y] = 0$
homogeneous DE
Sol: gen sol
Particular solution etc.

$y_2(t) \rightarrow$ Sol of $L[y] = 1$
Sol: gen sol
Particular sol: